Retail Customer & Sales Intelligence Analysis

Oracle SQL*Plus — Query and Expected Output Reference

This guide contains 10 analysis queries and example result tables computed from the supplied cleaned_sales dataset. The tables show the expected result format and values; they are not screenshots or logs from Oracle SQL*Plus. Actual output may vary slightly with table-import types or settings.

Before running: Import the existing cleaned_sales.csv into Oracle as a table named CLEANED_SALES. Then execute each query in the .sql file. These queries are written for Oracle 10g (ROWNUM used for top-10 queries).

01. Overall sales KPIs

```
SELECT SUM(Revenue) AS Total_Sales,
       SUM(Quantity) AS Total_Quantity,
       COUNT(DISTINCT InvoiceNo) AS Total_Orders,
       COUNT(DISTINCT CustomerID) AS Total_Customers,
       COUNT(DISTINCT StockCode) AS Total_Products
FROM cleaned_sales;
```

Expected output format

TOTAL_SALES	TOTAL_QUANTITY	TOTAL_ORDERS	TOTAL_CUSTOMERS	TOTAL_PRODUCTS
10,666,684.54	5,588,376	19,960	4,338	3,922

02. Monthly revenue trend

```
SELECT YearMonth, ROUND(SUM(Revenue), 2) AS Monthly_Revenue
FROM cleaned_sales
GROUP BY YearMonth
ORDER BY YearMonth;
```

Expected output format

YEARMONTH	MONTHLY_REVENUE
2010-12	823,746.14
2011-01	691,364.56
2011-02	523,631.89
2011-03	717,639.36
2011-04	537,808.62
2011-05	770,536.02
2011-06	761,739.90
2011-07	719,221.19
2011-08	759,138.38
2011-09	1,058,590.17
2011-10	1,154,979.30
2011-11	1,509,496.33
2011-12	638,792.68

03. Top 10 countries by revenue (Oracle 10g)

```
SELECT Country, Total_Revenue
FROM (
    SELECT Country, ROUND(SUM(Revenue), 2) AS Total_Revenue
    FROM cleaned_sales
    GROUP BY Country
    ORDER BY SUM(Revenue) DESC
)
WHERE ROWNUM <= 10;
```

Expected output format

COUNTRY	TOTAL_REVENUE
United Kingdom	9,025,222.08
Netherlands	285,446.34
EIRE	283,453.96
Germany	228,867.14
France	209,715.11
Australia	138,521.31
Spain	61,577.11
Switzerland	57,089.90
Belgium	41,196.34
Sweden	38,378.33

04. Top 10 products by revenue (Oracle 10g)

```
SELECT StockCode, Description, Total_Revenue
FROM (
    SELECT StockCode, Description, ROUND(SUM(Revenue), 2) AS Total_Revenue
    FROM cleaned_sales
    GROUP BY StockCode, Description
    ORDER BY SUM(Revenue) DESC
)
WHERE ROWNUM <= 10;
```

Expected output format

STOCKCODE	DESCRIPTION	TOTAL_REVENUE
DOT	DOTCOM POSTAGE	206,248.77
22423	REGENCY CAKESTAND 3 TIER	174,484.74
23843	PAPER CRAFT , LITTLE BIRDIE	168,469.60
85123A	WHITE HANGING HEART T-LIGHT HOLDER	104,340.29
47566	PARTY BUNTING	99,504.33
85099B	JUMBO BAG RED RETROSPOT	94,340.05
23166	MEDIUM CERAMIC TOP STORAGE JAR	81,700.92
M	Manual	78,110.27
POST	POSTAGE	78,101.88
23084	RABBIT NIGHT LIGHT	66,964.99

05. Top 10 customers by revenue (Oracle 10g)

```
SELECT CustomerID, Total_Revenue, Total_Orders, Total_Quantity
FROM (
    SELECT CustomerID,
           ROUND(SUM(Revenue), 2) AS Total_Revenue,
           COUNT(DISTINCT InvoiceNo) AS Total_Orders,
           SUM(Quantity) AS Total_Quantity
    FROM cleaned_sales
    WHERE CustomerID IS NOT NULL
    GROUP BY CustomerID
    ORDER BY SUM(Revenue) DESC
)
WHERE ROWNUM <= 10;
```

Expected output format

CUSTOMERID	TOTAL_REVENUE	TOTAL_ORDERS	TOTAL_QUANTITY
14646	280,206.02	73	196,915
18102	259,657.30	60	64,124
17450	194,550.79	46	69,993
16446	168,472.50	2	80,997
14911	143,825.06	201	80,265
12415	124,914.53	21	77,374
14156	117,379.63	55	57,885
17511	91,062.38	31	64,549
16029	81,024.84	63	40,208
12346	77,183.60	1	74,215

06. Average order value

```
SELECT ROUND(AVG(Order_Revenue), 2) AS Average_Order_Value
FROM (
    SELECT InvoiceNo, SUM(Revenue) AS Order_Revenue
    FROM cleaned_sales
    GROUP BY InvoiceNo
);
```

Expected output format

AVERAGE_ORDER_VALUE
534.40

07. Revenue and quantity by country (top 10 quantity)

```
SELECT Country, Total_Quantity
FROM (
    SELECT Country, SUM(Quantity) AS Total_Quantity
    FROM cleaned_sales
    GROUP BY Country
    ORDER BY SUM(Quantity) DESC
)
WHERE ROWNUM <= 10;
```

Expected output format

COUNTRY	TOTAL_QUANTITY
United Kingdom	4,662,390
Netherlands	200,361
EIRE	147,173
Germany	119,261
France	112,103
Australia	83,901
Sweden	36,083
Switzerland	30,629
Spain	27,940
Japan	26,016

08. Cancellation vs non-cancellation summary

```
SELECT IsCancellation,
       COUNT(*) AS Row_Count,
       ROUND(SUM(Revenue), 2) AS Total_Revenue
FROM cleaned_sales
GROUP BY IsCancellation
ORDER BY IsCancellation;
```

Expected output format

ISCANCELLATION	ROW_COUNT	TOTAL_REVENUE
False	530,104	10,666,684.54
True	0	0.00

09. Missing customer IDs

```
SELECT COUNT(*) AS Missing_CustomerID_Rows
FROM cleaned_sales
WHERE CustomerID IS NULL;
```

Expected output format

MISSING_CUSTOMERID_ROWS
132,220

10. Monthly order count

```
SELECT YearMonth, Total_Orders
FROM (
    SELECT YearMonth, COUNT(DISTINCT InvoiceNo) AS Total_Orders
    FROM cleaned_sales
    GROUP BY YearMonth
    ORDER BY YearMonth
);
```

Expected output format

YEARMONTH	TOTAL_ORDERS
2010-12	1,559
2011-01	1,086
2011-02	1,100
2011-03	1,454
2011-04	1,246
2011-05	1,681
2011-06	1,533
2011-07	1,475
2011-08	1,361
2011-09	1,837
2011-10	2,040
2011-11	2,769
2011-12	819